Rank #1 Science Classes by Doctors, Engineers and Textbook Authors
SHIV SCIENCE CLASSES
11th & 12th Science | MHTCET | NEET | JEE | NDA | NATA

Class: 12th[ENTRANCE] **MHCET PAPER - PCM** **Total Marks :200**

1. The moment of inertia of a solid sphere of mass M and radius R about its diameter is:
(A) $(2/5)MR^2$ (B) $(1/2)MR^2$
(C) $(2/3)MR^2$ (D) MR^2

2. The SI unit of angular momentum is:

(A) $kg \cdot m \cdot s^{-1}$ (B) $kg \cdot m^2 \cdot s^{-1}$
(C) $kg \cdot m^2 \cdot s^{-2}$ (D) $N \cdot m^{-1}$

3. A wheel rotates at 10 $rad \cdot s^{-1}$. If its angular velocity becomes 30 $rad \cdot s^{-1}$ in 5 seconds under constant angular acceleration, the angular acceleration is:

(A) 2 $rad \cdot s^{-2}$ (B) 4 $rad \cdot s^{-2}$
(C) 6 $rad \cdot s^{-2}$ (D) 8 $rad \cdot s^{-2}$

4. A solid cylinder of mass 2 kg and radius 0.5 m rotates about its central axis at 4 $rad \cdot s^{-1}$. Its rotational kinetic energy is:

(A) 0.5 J (B) 1 J (C) 2 J (D) 4 J

5. A uniform disc of mass M and radius R rotates freely at angular velocity ω. A small mass M is gently placed at its rim and sticks to it. The final angular velocity is:

(A) ω/2 (B) ω/3 (C) 2ω/3 (D) 3ω/2

6. The pressure at a depth h in a liquid of density ρ, due to the liquid alone, is:

(A) ρgh (B) ρg/h (C) gh/ρ (D) ρh/g

7. According to Bernoulli's principle, where the speed of an ideal fluid increases along a horizontal streamline, its static pressure generally:
(A) Increases (B) Decreases
(C) Remains unchanged (D) Becomes zero

8. Water flows through a pipe whose cross-sectional area decreases from 4 cm^2 to 2 cm^2. If its speed in the wider section is 3 $m \cdot s^{-1}$, the speed in the narrower section is:
(A) 1.5 $m \cdot s^{-1}$ (B) 3 $m \cdot s^{-1}$
(C) 6 $m \cdot s^{-1}$ (D) 12 $m \cdot s^{-1}$

9. The average translational kinetic energy per molecule of an ideal gas at absolute temperature T is:
(A) kT (B) (1/2)kT
(C) (3/2)kT (D) (5/2)kT

10. The absolute temperature of an ideal gas is increased from 300 K to 1200 K. The ratio of its new to original root-mean-square speed is:
(A) 2 (B) 4
(C) 1/2 (D) 1/4

11. For an ideal gas undergoing an isothermal process, the change in internal energy is: (A) Positive (B) Negative (C) Zero (D) Infinite

12. A gas absorbs 500 J of heat and performs 200 J of work on its surroundings. Its change in internal energy is: (A) 700 J (B) 300 J (C) −300 J (D) −700 J

13. The time period of a simple pendulum of length L, for small oscillations, is:

(A) $2\pi\sqrt{(L/g)}$ (B) $2\pi\sqrt{(g/L)}$ (C) $2\pi L/g$ (D) $\sqrt{(Lg)}$

14. A particle executes SHM with amplitude A = 0.10 m and angular frequency ω = 5 $rad \cdot s^{-1}$. Its maximum speed is:

(A) 0.02 $m \cdot s^{-1}$ (B) 0.25 $m \cdot s^{-1}$
(C) 0.50 $m \cdot s^{-1}$ (D) 2.0 $m \cdot s^{-1}$

15. In SHM, the displacement is x = A/2. The magnitude of acceleration is:

(A) $\omega^2 A/2$ (B) $\omega^2 A$
(C) $2\omega^2 A$ (D) Zero

16. Two coherent waves of equal amplitude arrive in phase at a point. The resulting amplitude is:

(A) Zero (B) Equal to either wave's amplitude
(C) Twice either wave's amplitude
(D) Four times either wave's amplitude

17. The beat frequency produced by two sound waves of frequencies 256 Hz and 260 Hz is:

(A) 2 Hz (B) 4 Hz
(C) 258 Hz (D) 516 Hz

18. A stationary wave is formed by two waves travelling in opposite directions, each of wavelength 0.8 m. The distance between two consecutive nodes is:

(A) 0.2 m (B) 0.4 m
(C) 0.8 m (D) 1.6 m

19. In Young's double-slit experiment, the fringe width is:

(A) $\lambda d/D$ (B) $\lambda D/d$
(C) dD/λ (D) $D/(\lambda d)$

20. In Young's double-slit experiment, $\lambda = 600$ nm, the screen distance is $D = 1.5$ m, and the slit separation is $d = 0.5$ mm. The fringe width is: (A) 0.18 mm (B) 0.9 mm (C) 1.8 mm (D) 3.6 mm

21. For a single slit of width a, the first minimum in diffraction occurs at:
(A) $a \sin\theta = \lambda$ (B) $a \cos\theta = \lambda$
(C) $a \tan\theta = 2\lambda$ (D) $a \sin\theta = 2\lambda$

22. The electric field inside a conductor in electrostatic equilibrium is: (A) Zero
(B) Infinite (C) Uniform but non-zero
(D) Dependent on its mass

23. The SI unit of electric potential is:

(A) Coulomb (B) Volt (C) Tesla (D) Farad

24. Two point charges +2 μC and +3 μC are separated by 0.30 m in vacuum. Using $k = 9 \times 10^9\ N\cdot m^2\cdot C^{-2}$, the magnitude of the electrostatic force is:

(A) 0.2 N (B) 0.6 N (C) 1.2 N (D) 6.0 N

25. A 4 μF capacitor is connected to a 12 V battery. The charge stored is:
(A) 3 μC (B) 16 μC (C) 48 μC (D) 96 μC

26. Two capacitors of 3 μF and 6 μF are connected in series across a 12 V battery. The potential difference across the 3 μF capacitor is:

(A) 4 V (B) 6 V (C) 8 V (D) 10 V

27. Ohm's law at constant physical conditions is:

(A) $V = IR$ (B) $V = I/R$
(C) $R = VI$ (D) $I = VR$

28. Two resistors of 4 Ω and 6 Ω are connected in series. Their equivalent resistance is:

(A) 2 Ω (B) 2.4 Ω
(C) 10 Ω (D) 24 Ω

29. A 12 V battery is connected across a 4 Ω resistor. The current is:

(A) 0.33 A (B) 2 A (C) 3 A (D) 48 A

30. The magnetic field at the centre of a circular loop of radius R, carrying current I, with N turns, is:

(A) $\mu_0 NI/(2R)$ (B) $\mu_0 NI/R$
(C) $\mu_0 I/(4\pi R)$ (D) $\mu_0 NR/(2I)$

31. A straight conductor of length 0.5 m carries a current of 4 A perpendicular to a uniform magnetic field of 0.2 T. The magnetic force is:
(A) 0.04 N (B) 0.2 N
(C) 0.4 N (D) 2.0 N

32. A material that is weakly repelled by a magnetic field is:
(A) Paramagnetic (B) Ferromagnetic
(C) Diamagnetic (D) Ferrimagnetic

33. According to Faraday's law, the magnitude of induced emf in a coil of N turns is:
(A) $N\,|d\Phi^B/dt|$ (B) $\Phi^B/(Nt)$
(C) $N\Phi^B t$ (D) dN/dt

34. The magnetic flux through a single loop decreases uniformly from 0.8 Wb to 0.2 Wb in 0.3 s. The magnitude of induced emf is:

(A) 0.2 V (B) 0.5 V
(C) 2.0 V (D) 3.0 V

35. A conducting rod of length 0.5 m moves at 8 $m\cdot s^{-1}$ perpendicular to a magnetic field of 0.4 T. If the circuit resistance is 2 Ω, the induced current is: (A) 0.2 A (B) 0.4 A (C) 0.8 A
(D) 1.6 A

36. The RMS value of an alternating voltage with peak value V_0 is: (A) V_0
(B) $V_0/\sqrt{2}$ (C) $\sqrt{2}V_0$
(D) $V_0/2$

37. An ideal inductor of inductance 0.2 H is connected to a 50 Hz AC supply. Its inductive reactance is approximately: (A) 15.7 Ω
(B) 31.4 Ω (C) 62.8 Ω
(D) 100 Ω

38. In a series RLC circuit at resonance, the impedance is:

(A) Zero (B) Equal to the resistance
(C) Equal to the inductive reactance (D) Infinite

39. A series RLC circuit has $R = 30\ \Omega$, $X^L = 50\ \Omega$, and $X^c = 10\ \Omega$. Its impedance is:

(A) 40 Ω (B) 50 Ω

(C) 60 Ω (D) 90 Ω

40. The energy of a photon of frequency ν is:

(A) h/ν (B) hν
(C) hν² (D) ν/h

41. The work function of a metal is 2 eV, and incident photons have energy 5 eV. The maximum kinetic energy of emitted photoelectrons is: (A) 2 eV
(B) 3 eV (C) 5 eV
(D) 7 eV

42. The number of neutrons in a nucleus with mass number A and atomic number Z is:
(A) A + Z (B) Z – A
(C) A – Z (D) AZ

43. A radioactive sample has a half-life of 10 days. The fraction of the original sample remaining after 30 days is:
(A) 1/2 (B) 1/4 (C) 1/6 (D) 1/8

44. The radius of a nucleus is proportional to $A^{(1/3)}$. If the mass number increases from 8 to 64, the nuclear radius becomes:

(A) Twice as large (B) Four times as large
(C) Eight times as large (D) Unchanged

45. In an intrinsic semiconductor, the number of free electrons is:

(A) Greater than the number of holes
(B) Less than the number of holes
(C) Equal to the number of holes
(D) Always zero

46. A p-n junction diode is forward biased when:

(A) The p-side is connected to the negative terminal and the n-side to the positive terminal
(B) Both sides are connected to the positive terminal
(C) The p-side is connected to the positive terminal and the n-side to the negative terminal
(D) Both sides are connected to the negative terminal

47. The primary function of a rectifier is to convert:

(A) DC into AC (B) AC into DC
(C) Heat into electricity (D) Light into sound

48.*J* For an ideal common-emitter transistor, the current gain is β = 100. If the base current is 20 μA, the collector current is approximately:

(A) 0.2 mA (B) 2 mA
(C) 20 mA (D) 200 mA

49. A Zener diode is used as a voltage regulator with a 5.6 V Zener voltage. Assuming it remains in breakdown and has sufficient current, the approximate load voltage is:

(A) 0 V (B) 2.8 V (C) 5.6 V
(D) Equal to the supply voltage under all conditions

50. The majority charge carriers in an n-type semiconductor are:

(A) Holes (B) Electrons
(C) Protons (D) Positive ions

---------- Chemistry ----------

51. What happens in Schottky defect?

A) The density of crystal increases.

B) Unequal number of cations and anions are missing from the lattice.

C) Either cation or anion leaves its normal site and occupies interstitial site.

D) Equal number of cations and anions are missing from their regular positions in the crystal lattice.

52. For a certain reaction, change in enthalpy is $-225\ KJ$ *and entropy change* $= -150\ JK^{-1}$

Find the temperature so that ΔG is zero.

A) 350 K B) $1500\ KJ$

C) $105\ KJ$ D) $1800\ KJ$

53. What is the pH of $0.01\ M$ *acetic acid that is* 1% ionized ?

A) 1 B) 2

C) 3 D) 4

54. Identify the **FALSE** statement from the following.

A) Standard cell potential of galvanic cell is an extensive property.

B) Standard electrode potential is an intensive property.

C) For spontaneous cell reaction in galvanic cell, free energy change should be negative.

D) For spontaneous cell reaction in galvanic cell equilibrium constant of reaction should greater than 1

55. Find the percentage of unreacted reactant for a zero order reaction in 90 seconds having rate constant 1 mol dm^{-3} s^{-1} & initial concentration 100 mol dm^{-3}.

A) 5 % B) 10%

C) 15% D) 20%

56. Which of the following factors is responsible for the anomalous behavior of oxygen?
(A). small atomic size.
(B). higher electronegativity.
(C). low O - O bond dissociation energy.
(D). absence of d - orbital
(E). presence of vaccant d - orbitals.
Choose the **correct** answer from the options given below:

A) (A), (B), (C) and (E) only. B) (A), (B) and (D) only.
C) (A), (B), (C) and (D). D) (B), (C) and (D) only.

57. Identify the number of lanthanoids among the following, exhibiting the $+4$
Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy

A) 5 B) 3 C) 4 D) 6

58. Identify the isomerism exhibited by complex compounds $[Co(NH_3)_5SO_4]Br[Co(NH_3)_5Br]SO_4$

A) Coordination isomers B) Geometrical isomers

C) Optical isomers D) Ionisation isomers

59. $CH_3\text{-}CH_2\text{-}Cl + NaI \xrightarrow{\text{acetone}} CH_3\text{-}CH_2\text{-}I + NaCl \downarrow$

A) Swartz reaction B) Finkelstein reaction

C) Wurtz reaction D) Sandmeyer reaction

60. An organic compound gives hydrogen on reaction with sodium metal and it also forms an aldehyde having molecular formula C_2H_4O on oxidation with pyridinium chlorochromate, identify the organic compound from following ?

A) Acetone B) Ethyl alcohol

C)Dimethyl ether D)Acetic acid

61. When silver crystallizes, it forms face centered cubic cells, if the volume of unit cells 6.84×10^{-23} cm^3 $N_A = 6.022 \times 10^{23}$

A) 12.49 gm cm^{-3} B) 10.49 gm cm^{-3}

C) 16.89 gm cm^{-3} D) 20.49 gm cm^{-3}

62. For the cell: $Ni_{(s)}|Ni^{++}_{(1M)}||Cu^{++}_{(1M)}|Cu_{(s)}$ if the concentration of an ion in the anode compartment is doubled then the emf of cell will be:

A) increased by 2 B) decreased by 2

C) increased by 0.00891V D) decreased by$0.00891\ V$

63. Which of the following elements exhibits only one value of oxidation state?

A) Mn B) Fe C) Zn D) W

64. Which of the following compounds cannot be prepared by Williamson's synthesis?

A) $C_6H_5OCH_2CH_3$ B) $CH_3CH_2O\ CH_2CH_3$

C) $(CH3)_3C\text{-}O\text{-}CH_2CH_3$ D) $(CH3)_3C\text{-}O\text{-}C(CH_3)_3$

65. Which of the following amines exhibits the Carbylamine test?

A) Primary amine B) Secondary amine

C) Tertiary amine D) Quaternary ammonium salt

66. Which of the following has highest pK_b value?

A) $CCl_3 - CH_2 - NH_2$

B) $CCl_3 - CH_2 - CH_2 - NH_2$

C) $CF_3 - CH_2 - NH_2$

D) $CF_3 - CH_2 - CH_2 - NH_2$

67. What is the mass of 5 unit cells of sodium element having bcc structure
(molar mass of Na $= 23$ g/ mol)

A) 1.05×10^{-22} g B) 4.123×10^{-23} g

C) 7.638×10^{-23} g D) 3.819×10^{-22} g

68. Calculate the osmotic pressure of a solution containing 3.42 g 180 g/mol in1 litre of water at 400 K

A) 0.624 atm B) 0.180 atm

C) 0.340 atm D) 0.420 atm

69. Which of following is the colligative property of a solution?

A) Vapour pressure B) Boiling point

C) Osmotic pressure D) Freezing point

70. Find the number of allotropic forms of Tellurium under normal conditions.

A) 2 B) 3 C) 6 D) 8

71. 9 gram anhydrous oxalic acid (mol. Wt. = 90) was dissolved in 9.9 moles of water. If vapour pressure of pure water is $P_1^o c$ the vapour pressure of solution is

A) $0.99P^o_1$ B) $0.1P^o_1$

C) $0.91P^o_1$ D) $1.1P^o_1$

72. Identify the major product obtained when chlorobenzene is reacted with chlorine in the presence of anhydrous $FeCl_3$

A) 1,2-Dichlorobenzene B) Benzene hexachloride

C) 1,4-Dichlorobenzene D) 2,4,6-Trichlorobenzene

73. Calculate the amount of heat evolved when $33CO_2(g)$ formed if the reaction is

$CH_2O(g) + O_2(g) \rightarrow CO_2(g) + H_2O(g);\ \Delta H = -527$

A) $527\ KJ$ B) 395.5 KJ
C) $263.5\ KJ$ D) $131.8\ KJ$

74. Identify the position of lanthanum in terms of the following blocks in the modern periodic table?

A) f block B) p block C) s block D) d block

75. Identify ' B′ in the following series of reactions

$$\text{Ethanol} \xrightarrow[H_2SO_4,\Delta]{NaBr} A \xrightarrow[\text{Dry ether}]{Mg} B$$

A) Ethyl magnesium bromide B) Ethyl bromide

C) Sodium ethoxide D) Ethene

76. What is the effective atomic number of Cu in $\left[Cu(NH_3)_4\right]^{2+}$

A) 18 B) 33 C) 35 D) 36

77. Identify the product ' B ' in the following reaction.

$$\text{Ethylphenyl ketone} \xrightarrow[-H_2O]{H_2N-NH_2} A \xrightarrow{KOH,HO_1(CH_2)_2OH} B$$

A) Phenylhydrazone B) Ethylbenzene

C) n-propylbenzene D) Isopropylbenzene

78. Which of the following compounds when comes in contact with the eyes, damages the cornea?

A) Chloromethane B) Dichloromethane

C) Trichloromethane D) Tetrachloromethane

79. What type of amines exhibits diazotization reaction?

A) Aliphatic primary amine

B) Aromatic primary amine

C) Aliphatic secondary amine

D) Aromatic secondary amine

80. For a first order reaction, the half-lifeB period is

A) dependent on the square of the initial concentration
B) dependent on first power of initial
C) dependent on the square root of initial concentration
D) independent on initial concentration

81. Which of the following salts does not undergo hydrolysis?

A) Ammonium chloride. B) Sodium carbonate.

C) Ammonium cyanide. D) Potassium nitrate.

82. $E°cell = 1 \cdot 049$ V and involves transfer of 2 electrons, calculate equilibrium constant of cell?

A) 2.75×10^{35} B) 2.75×10^{10}

C) $0{\cdot}524\times10^{35}$ D) $2{\cdot}098\times10^{10}$

83. Identify the correct decreasing order of thermal stability from following.

A) ClF >ICl > IBr> BrCl B) BrCl > IBr > ICl >ClF

C) ClF > IBr> BrCl> ICl D) ClF > BrCl > IBr > ClF

84. Which of the following elements does not belong to Lanthanoids?

A) Nd B) Lu C) Pa D) Pr

85. For first order reaction the concentration of reactant decreases form $0 \cdot 2\ to\ 0.1M$ in 100 minutes. What is the rate constant of the reaction?

A) 6.93 min^{-1} B) $69{\cdot}3\ min^{-1}$

C) $6.93\times10^{-3}\ min^{-1}$ D) $144{\cdot}3\ min^{-1}$

86. Which among the following compounds contains highest number of chlorine atoms in their single molecule?

A) Phosgene B) Tear gas

C) Mustard gas D) Phosphine

87. Identify geometry of $\left[Ni(Cl)_4\right]^{2-}$

A) Square planar B) Tetrahedral

C) Octahedral D) Trigonal bipyramidal

88. Which of the following is NOT a disaccharide ?

A) Cane sugar B) Raffinose

C) Lactose D) Maltose

89. In an isothermal and reversible process, $1{\cdot}6\times10^{-2}$ kg O_2 expands from 10 dm^3 to 100 dm^3 at 300 K , work done in the process is (R = 8.314 J)

A) -1436 J B) −5744 J

C) -4308 J D) -2872 J

90. Calculate solubility ($moldm^{-3}$) of a sparingly soluble electrolyte AB at 298 K if its solubility product is 1.6×10^{-5} ?

A) 1.6×10^{-3} B) 2.5×10^{-3}

C) 4.0×10^{-3} D) 8.0×10^{-3}

91 Which from following statements regarding mercury battery is NOT true?

A) It is a primary dry cell.
B) It consists of Zn anode amalgamated with mercury.
C) The electrolyte is strongly acidic.
D) In this Hg is obtained by reduction of HgO

92.. Carbolic acid is oxidised by acidified sodium dichromate to give.

A) Acetone B) Ethyl methyl ketone

C) Benzoquinone D) Anthraquinone

93. Which of the following polymers belongs elastomers?

A)Polystyrene B)Polythene

C) Polyvinyls D) Neoprene

94. Quaternary ammonium salts are used in ___

A) Anionic detergents. B) Cationic detergents.

C) Nonionic detergents. D)Soaps.

95. In haloform reaction of acetaldehyde and all methyl ketones (CH_3COR), methyl group is converted to

A) Sodium salt B) Haloform

C) Sodium hydroxide D) Sodium hypohalite

96. Which of the following nitrogen bases is absent in RNA?

A) Thymine B)Cytosine C)Uracil D) Adenine

97. What is the number of chiral carbon atoms in the open chain structure of glucose and fructose respectively?

A) 4 carbon atoms in each B) 3 carbonatoms in each
C) 4 carbon atoms in glucose and 3 carbon atoms in fructose
D) 3 carbon atoms in glucose and 4 carbon atoms in fructose

98. When 2 moles of an ideal gas are expanded isothermally from a volume of 2.5 L to 15.0 L against constant external pressure of 760 mmHg Calculate the amount of work done in joule?

A) −253·25 J B) −190·0 J

C) −1924·0 J D) −25·325 J

99. What is the product formed when cumene is air oxidized in presence of co-naphthenate and further treated with dilute acid?

A) Cumene hydroperoxide B) Phenol + Acetone

C) Acetone and Benzoic acid D) Phenol and CO_2

100. Three moles of an ideal gas are expanded isothermally from a volume of $300cm^3$ *to* 2.5L at 300K against a pressure of 1.9atm. The work done in joules is

A) −423.56 J B) +423.56J

C) −4.18J D) +4.8J

101. The negation of p → q is equivalent to:

(A) ~p ∨ q (B) p ∧ ~q
(C) ~p ∧ q (D) p ∨ ~q

102. The proposition p ∨ ~p is a:

(A) Contradiction (B) Contingency
(C) Tautology (D) Converse

103. By De Morgan's law, ¬(p ∧ q) is equivalent to:
(A) ~p ∧ ~q (B) p ∨ q
(C) ~p ∨ ~q (D) p ∧ ~q

104. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, then which of the following is not true?

(A) $A^{-1}=A^2$
(B) A^2 is a triangular matrix.
(C) A is a non-singular matrix.
(D) A^3 is an identity matrix.

105. The sum of the matrix $\begin{bmatrix} 2 & -3 \\ 5 & -7 \end{bmatrix}$ and its multiplicative inverse is

(A) $\begin{bmatrix} 2 & -3 \\ 5 & -7 \end{bmatrix}$ B) $\begin{bmatrix} -7 & 3 \\ -5 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

106. If A is a square matrix and $A^T = A$, then A is:
(A) Skew-symmetric (B) Symmetric
(C) Singular (D) Identity

107. *If* $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$, *then* $A^{-1} =$

(A) $\frac{1}{6}$ (A−4I) (B) $\frac{1}{6}$(A+5I)
(C) $\frac{1}{6}$ (A+4I) (D) $\frac{1}{6}$ (A−5I)

108. The value of sin 30° + cos 60° is:

(A) 0 (B) ½ (C) 1 (D) √3

109. If θ is acute and tan θ = 3/4, then sin θ equals:

(A) 3/5 (B) 4/5 (C) 3/4 (D) 4/3

110. The solutions of $2\sin^2 x - 1 = 0$ in $0 \le x < 2\pi$ are:

(A) π/4, 3π/4 only (B) π/4, 3π/4, 5π/4, 7π/4
(C) π/2, 3π/2 (D) 0, π

111. The equation $x^2 - y^2 = 0$ represents the pair of lines:

(A) y = x and y = −x (B) y = 2x and y = −2x
(C) x = 0 and y = 0 (D) y = x and y = 0

112. The point of intersection of x + y = 4 and x − y = 2 is:

(A) (2, 2) (B) (3, 1) (C) (1, 3) (D) (4, 2)

113. The pair of lines $x^2 - 5xy + 6y^2 = 0$ has slopes:

(A) 2 and 3 (B) 1/2 and 1/3
(C) −2 and −3 (D) 3 and −2

114. If a = 2i + 3j, then |a| is:

(A) √5 (B) 5 (C) √13 (D) 13

115. *If the vector $\bar{r}$ is perpendicular to $\hat{\imath} - \hat{k}$ & $\bar{r} \times (\hat{\imath} + x\hat{\jmath} - \hat{k}) = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}\bar{r} \cdot$ then the value of $\bar{r} \cdot (\hat{\imath} + x\hat{\jmath} - \hat{k}) =$*

(A) -2 (B) -1 (C) 1 (D) -2

116. If a = i + 2j and b = 3i − j, then a · b equals:

(A) 1 (B) 3 (C) 5 (D) −1

117. The angle between vectors a and b is 60°. If |a| = 2 and |b| = 4, then a · b is:

(A) 2 (B) 4 (C) 6 (D) 8

118. A direction vector of the line through (1, 2, 3) and (3, 2, 5) is:

(A) (2, 0, 2) (B) (4, 4, 8) (C) (1, 0, 1) (D) (2, 2, 2)

119. The distance between (0, 0, 0) and (2, 3, 6) is:

(A) 7 (B) 6 (C) 5 (D) 9

120. The plane with normal vector (2, −1, 3) passing through (1, 0, 2) has equation:

(A) 2x − y + 3z − 8 = 0
(B) 2x + y + 3z − 8 = 0
(C) 2x − y + 3z + 8 = 0
(D) x − 2y + 3z − 8 = 0

121. The distance of point (1, 2, 2) from plane x + 2y + 2z − 10 = 0 is:

(A) 1/3 (B) 1 (C) 2 (D) 3

122. In a linear programming problem, the feasible region is the set of points that:

(A) Satisfy all constraints, including non-negativity
(B) Maximize every variable separately
(C) Make the objective function zero only
(D) Violate at least one constraint

123. For x ≥ 0, y ≥ 0, x + y ≤ 4, and x ≤ 2, the point that maximizes Z = 3x + 2y is:

(A) (0, 4) (B) (2, 2) (C) (2, 0) (D) (0, 0)

124. The corner-point method for a bounded linear programming problem evaluates the objective function at:

(A) Only the origin
(B) Every point in the feasible region
(C) The vertices of the feasible region
(D) Only points where x = y

125. *[*Maximize Z = 5x + 4y subject to x + y ≤ 5, 2x + y ≤ 8, x ≥ 0, y ≥ 0. The maximum value is:

(A) 20 (B) 21 (C) 23 (D) 25

126. If $y = \cos^{-1}(\sin x)\frac{\pi}{2} < x < \pi \frac{dy}{dx} = \ldots$

(A) -1 (B) 0 (C) 1 (D) 2A

127. If $y = \cos^{-1} x \frac{d^2y}{dx^2} =$

(A) $\sec^2 y \cdot \tan y$ (B) $-\sec^2 y \cdot \tan y$
(C) $\operatorname{cosec}^2 y \cdot \cot y$ (D) $-\operatorname{cosec}^2 y \cdot \cot y$

128. If y = log (x² + 1), then dy/dx is:

(A) 1/(x² + 1) (B) 2x/(x² + 1)
(C) 2/(x² + 1) (D) 2x log(x² + 1)

129. The slope of the tangent to $y = x^2$ at x = 1 is:

(A) 1 (B) 2 (C) 3 (D) 4

130. The stationary points of $f(x) = x^3 - 3x$ occur at:

(A) x = 0 only (B) x = ±1
(C) x = ±3 (D) x = 1 only

131. A rectangle has perimeter 20 cm. Its maximum possible area is:

(A) 20 cm² (B) 24 cm² (C) 25 cm² (D) 50 cm²

132. Evaluate $\int e^x (\sin x + 2\cos x) \sin x \; dx$

(A) $e^x \cos x + C$ (B) $e^x \sin x + C$
(C) $e^x \sin 2x + C$ (D) $e^x \sin 2x + C$

133. $\int f(x)\,dx = g(x) + c\int x\, f(x^2)\,dx =$

(A) $2g(x^2)+c$ (B) $2\,g(x) + c$
(C) $\frac{1}{2}g(x^2)+c$ (D) $\frac{1}{2}\,g(x) + c$

134. $\int \frac{4x+5}{2x+1}\,dx$ *equals to*

(A) $2x + \frac{3}{2}\log|2x+1|+c$ (B) $2x-\frac{3}{2}\log|2x+1| + c$
(C) $x + \frac{3}{2}\log|2x+1| + c$ (D) $x - \frac{3}{2}\log|2x+1| + c$

135. Evaluate

(A)$\sin^{-1}(x-1)+c$ (B) $\cos^{-1}(x-1) + c$
(C) $\tan^{-1}(x-1)+c$ (D) $\sin^{-1}x + c$

136. The area bounded by the parabola $y = 4x - x^2$ and the x-axis (in sq. units) is:

(A) $\frac{32}{3}$ (B) 8 (C) 12 (D) $\frac{16}{3}$

137. The value of the integral

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

138. If $\int \frac{1}{f(x)}\,dx = \log|f(x)|^2 + cf(x) = \cdots$

(A) $x + c$ (B) $\frac{x}{2} + c$ (C) $\frac{x^2}{2} + c$ (D) $x^2 + c$

139. The area under y = x from x = 0 to x = 2 and above the x-axis is:

(A) 1 square unit (B) 2 square units
(C) 3 square units (D) 4 square units

140. The area of the region bounded by the $yy = \cos x, y = \sin x\ 0 \le x \le \frac{\pi}{4}$

A) $\left(\sqrt{2}-1\right)$ sq. unit B) $2\left(\sqrt{2}-1\right)\ sq\ unit$

C) $\left(\sqrt{2}+1\right)\ sq.unit$ D) $\sqrt{2}\ sq.unit$

141. The general solution of $e^{\frac{dy}{dx}} = x$ is

(A) y = x logx – x + c (B) y = x log x + c
(C) y = 1/x + c (D) y = x.logx + x + c

142. The general solution of dy/dx = y is:

(A) y = x + C (B) $y = Ce^x$
(C) y = Cx (D) $y = e^x + C$

143. A body cools from $100°c\ to\ 80°c$ in 5 minutes in air at $20°c$. The time taken to cool from $80°C\ to\ 60°C\ will\ be$

(A) Less than 5 minutes (B) Equal to 5 minutes
(C) Greater than 5 minutes (D) Cannot be determined

144. A discrete random variable X takes values 1, 2, 3 with probabilities k, 2k, 3k. The value of k is:

(A) 1/3 (B) 1/6 (C) ½ (D) 1/9

145. For P(X = x) = x/6 for x = 1, 2, 3, the expected value E(X) is:

(A) 2 (B) 7/3 (C) 5/2 (D) 3

146. If X takes values 0 and 1 with P(X = 1) = 0.3, then E(X) is:

(A) 0 (B) 0.3 (C) 0.7 (D) 1

147. For a binomial distribution with parameters n and p, the mean is:

(A) np (B) nq (C) n/p (D) p/n

148. If X follows Binomial(n = 4, p = 0.5), then P(X = 2) is:

(A) ¼ (B) 3/8 (C) ½ (D) 5/8

149. If X follows Binomial (n = 6, p = 1/3), then P(X = 2) is:

(A) 40/243
(B) 80/243
(C) 16/81
(D) 10/27

150. /A binomial random variable has n = 5 and p = 0.2. The probability of zero successes is:

(A) 0.32768
(B) 0.4096
(C) 0.2
(D) 0.8